SolarGuard: Complete Wiring, Pin-to-Pin Netlist and Framework

Reference document, 10 October 2026. Companion to the four diagrams shown in the chat: system framework, power schematic, sensing circuits, actuation circuits. Everything below is a design to be verified on your actual modules (section 15).

1. How to read this document

- **Reference designators** name each part: U = module or IC, R = resistor, C = capacitor, D = diode, F = fuse, K = relay, S = switch, M = motor, P = pump, PV = solar cell.
- **Pin notation** is part.pin. Example: U6.GPIO1 means pin GPIO1 of the ESP32 (U6). A net is a set of pins joined by wire; every pin listed on one net line is connected together.
- **Net names:** GND, VBAT (battery rail), V5 (5 V rail), V3V3 (3.3 V from the ESP32 board).
- **Module pin names** differ between sellers. Match the function, not the position, and check the board's silkscreen.

2. System framework

Six layers, from energy to cloud:

1. **Energy layer:** Panel A feeds a charger, a 3-cell Li-ion pack with BMS, and the VBAT rail. Panel B is the untouched control and feeds a dummy load.
2. **Sensing layer:** panel voltage and current for both panels, three DS18B20 temperatures, DHT22 ambient, a reference solar cell for sunlight, a rain probe, a tank level sensor and two carriage end stops.
3. **Control layer:** ESP32-S2 (U6) reads sensors every 5 seconds and runs the state machine.
4. **Actuation layer:** relay K1 switches the pump, relay K2 moves Panel A between the charger and a dummy load (measure mode), and an L298N drives the carriage motor.
5. **Mechanical layer:** spray bar and nozzles at the top, two rods, carriage blocks, cross bar, wiper arms, squeegee, GT2 belt, motor and idler.
6. **Software layer:** firmware modules, Wi-Fi and NTP time, logging, dashboard.

Energy flow: PV A, F1, U4 (current sensor), K2, D1, U1 (charger), U2 (BMS and pack), S1, F3, VBAT. VBAT feeds U3 (buck, makes V5), the pump branch and the motor branch.

Signal flow: sensors to ESP32 analog and digital pins; ESP32 pins to relay and motor-driver inputs; ESP32 to Wi-Fi.

3. Reference designators

Power and protection

- PV1 Panel A (test), 12 V 20 W. PV2 Panel B (control), identical. PV3 reference mini cell (3 V, 150 mA).
- F1 and F2: 3 A fuses on the panel lines. F3: 5 A main fuse. F4: 2 A pump fuse. F5: 2 A motor-driver fuse.
- S1 master switch (rated at least 6 A DC).
- D1 blocking diode: SR560 or 1N5822 (3 A or more). D2 pump flyback diode: 1N4007.
- U1 charge controller with a Li-ion 3S (12.6 V) setting, or a CN3791 module set to 3S.
- U2 3S BMS. BT1, BT2, BT3: 18650 cells. J1: XT60 battery connector.
- U3 LM2596 buck module.
- RL1 and RL2: dummy loads, 15 ohm 25 W each.

Measurement and control

- U4 ACS712-30A for Panel A. U5 ACS712-30A for Panel B.
- U6 ESP32-S2-DevKitM-1.
- U7, U8, U9 DS18B20 probes: U7 on Panel A back, U8 on Panel B back, U9 in the tank or ambient shade.
- U10 DHT22. U11 rain sensor module (soil-moisture type). U12 tank level sensor. U13 top end-stop, U14 bottom end-stop (IR modules or push buttons).
- U15 two-channel 5 V relay module (K1 and K2). U16 L298N. M1 25GA-370 gear motor. P1 12 V pump.
- U17 optional thermal sensor (AMG8833 or MLX90614). U18 optional ESP32-CAM.

Passives

- R1, R3: 100k. R2, R4, R6, R8: 10k. R5, R7: 4.7k. R9: 10 ohm 0.5 W. R10: 4.7k. R11: 10k.
- C1: 220 uF 25 V. C2: 100 nF (across the motor). C3: 470 uF 10 V (on V5). C4, C5, C6, C7: 100 nF ceramic.
- Optional battery sensing: R12 100k, R13 22k, C8 100 nF (section 7).

4. Power architecture

Rails

- **VBAT:** the 3-cell pack, 9.0 V to 12.6 V (not a fixed 12 V). Pump and motor speed follow it.
- **V5:** 5.00 V from U3. Set it with the multimeter before connecting anything.
- **V3V3:** from the ESP32 board's regulator. Feeds sensors only (under 100 mA).
- **GND:** one common ground. Join all grounds at the battery negative (BMS P-) as a star point.

Current budget

- V5 loads: ESP32 about 0.12 A average (0.35 A peak), two relay coils up to 0.14 A, two ACS712 about 0.03 A, L298N logic about 0.04 A, sensors under 0.03 A. Total about 0.4 A (0.7 A peak). With the optional ESP32-CAM add about 0.2 A (0.5 A peak). U3 is rated 3 A.
- VBAT loads: pump 0.5 to 1 A, motor 0.3 to 0.8 A running (stall 1.5 to 2 A). Cleaning runs both, so budget about 1.5 A against F3 (5 A).
- Charging: Panel A gives up to about 1.3 A into 3 x 2600 mAh cells (0.5 C). Acceptable.
- Pack energy: 3 x 2.6 Ah x 3.7 V = about 28.9 Wh.

Wire gauges

- Panel, pack and VBAT wiring: 1.0 to 1.5 mm2 (18 to 16 AWG). Motor and pump: 0.75 to 1.0 mm2. Signals: 22 to 24 AWG.

Why the loads bypass the charger: the charger's LOAD terminals are not used. Its low-voltage cut-off is set for lead-acid, not for a 3S Li-ion pack, so loads run straight from the BMS output through S1 and F3.

5. Netlist: power

Panel A path

- **PVA0:** PV1(+), F1.in
- **PVA1:** F1.out, U4.IP+, R1.1 (voltage tap)
- **PVA2:** U4.IP-, K2.COM
- **PVA_CHG:** K2.NC, D1.anode
- **PVA_CHG2:** D1.cathode, U1.PV+
- **PVA_LOAD:** K2.NO, RL1.1

- **GND:** PV1(-), U1.PV-, U1.BAT-, RL1.2

On most PWM controllers PV- and BAT- are already common. If yours isolates them, join them.

Panel B path

- **PVB0:** PV2(+), F2.in
- **PVB1:** F2.out, U5.IP+, R3.1 (voltage tap)
- **PVB2:** U5.IP-, RL2.1
- **GND:** PV2(-), RL2.2

Battery and rail

- **CELL0:** BT1(-), U2.B-
- **CELL1:** BT1(+), BT2(-), U2.B1
- **CELL2:** BT2(+), BT3(-), U2.B2
- **PACK+:** BT3(+), U2.B+ (positive port), U1.BAT+, S1.in
- **PACK-:** U2.P- (negative port), U1.BAT-, the GND star point
- **SW:** S1.out, F3.in
- **VBAT:** F3.out, F4.in, F5.in, U3.IN+, R12.1 (if fitted)

BMS boards differ: if yours has a separate P+ pad, use P+ instead of B+ for the positive port.

Buck and 5 V rail

- **GND:** U3.IN-, U3.OUT-
- **V5:** U3.OUT+, U6.5V, U15.VCC, U4.VCC, U5.VCC, U16.+5V, C3(+), U18.5V (if fitted)
- **GND:** C3(-)

Pump branch

- **PUMP_FUSED:** F4.out, K1.COM
- **PUMP+:** K1.NO, P1(+), D2.cathode
- **GND:** P1(-), D2.anode (the anode joins P1(-) at the pump terminals; P1(-) returns to GND)
- K1.NC is not used.

Motor branch

- **MOT_FUSED:** F5.out, U16.+12V, C1(+)

- **GND:** U16.GND, C1(-)
- **M1_A:** U16.OUT1, M1.1, C2.1
- **M1_B:** U16.OUT2, M1.2, C2.2
- Remove the ENA and 5V-EN jumpers on U16. Feed its +5V pin from V5.
- Tie unused U16.IN3, IN4 and ENB to GND.

3.3 V rail

- **V3V3:** U6.3V3, U7.VDD, U8.VDD, U9.VDD, U10.VCC, U11.VCC, U12.VCC (if it works at 3.3 V), U13.VCC, U14.VCC, U17.VIN, R10.2, R11.2

6. Netlist: signals

Panel A voltage (VA)

- R1.1 to PVA1. R1.2, R2.1, C4.1, U6.GPIO1 join at node VA. R2.2 and C4.2 to GND.

Panel A current (IA)

- U4.GND to GND. U4.VCC to V5. U4.OUT to R5.1. R5.2, R6.1, C6.1, U6.GPIO2 join at node IA. R6.2 and C6.2 to GND.

Panel B voltage (VB) and current (IB)

- Same as A: R3, R4, C5 into U6.GPIO4; U5.OUT, R7, R8, C7 into U6.GPIO5.

Reference cell (VREF)

- PV3(+), R9.1, U6.GPIO7 join at one node. PV3(-) and R9.2 to GND.

Rain sensor

- U11.VCC to V3V3. U11.GND to GND. U11.AO to U6.GPIO6. U11.DO is not used.

DS18B20 temperatures (one shared 1-Wire bus)

- U7, U8, U9: VDD (red) to V3V3, GND (black) to GND, DQ (yellow or white) all to net DS_DQ.
- DS_DQ joins U6.GPIO10. R10 connects DS_DQ to V3V3 (4.7k pull-up).

DHT22

- 3-pin module: VCC to V3V3, GND to GND, DATA to U6.GPIO11 (pull-up is on the module).

- Bare 4-pin sensor: pin 1 VCC, pin 2 DATA to U6.GPIO11 with R11 (10k) to V3V3, pin 3 not connected, pin 4 GND.

Tank level sensor

- U12.VCC to V3V3 (check the sensor works at 3.3 V). U12.GND to GND. U12.OUT to U6.GPIO12.
- If it needs 5 V, power it from V5 and divide its output: 4.7k from OUT to the pin, 10k from the pin to GND.

End stops

- U13: VCC to V3V3, GND to GND, OUT to U6.GPIO13 (top stop).
- U14: VCC to V3V3, GND to GND, OUT to U6.GPIO14 (bottom stop).
- If using push buttons: one terminal to the GPIO, the other to GND, with INPUT_PULLUP.

Optional thermal sensor (I2C)

- U17.VIN to V3V3, GND to GND, SDA to U6.GPIO8, SCL to U6.GPIO9.

Relays and motor driver

- U15.VCC to V5. U15.GND to GND. U15.IN1 to U6.GPIO15. U15.IN2 to U6.GPIO16.
- K1.COM to PUMP_FUSED. K1.NO to PUMP+.
- K2.COM to PVA2. K2.NC to D1.anode. K2.NO to RL1.1.
- U16.ENA to U6.GPIO17. U16.IN1 to U6.GPIO18. U16.IN2 to U6.GPIO21.

7. ESP32-S2 master pin allocation

- **GPIO1:** input, ADC1_CH0, panel A voltage (VA).
- **GPIO2:** input, ADC1_CH1, panel A current (IA).
- **GPIO4:** input, ADC1_CH3, panel B voltage (VB).
- **GPIO5:** input, ADC1_CH4, panel B current (IB).
- **GPIO6:** input, ADC1_CH5, rain sensor analog out.
- **GPIO7:** input, ADC1_CH6, reference cell (VREF).
- **GPIO8, GPIO9:** I2C SDA and SCL for the optional thermal sensor.
- **GPIO10:** 1-Wire bus for the DS18B20 probes.
- **GPIO11:** DHT22 data.
- **GPIO12:** tank level sensor.
- **GPIO13, GPIO14:** top and bottom end stops.

- **GPIO15:** relay K1 (pump). **GPIO16:** relay K2 (measure mode).
- **GPIO17:** L298N ENA (PWM, 10 kHz, 8-bit). **GPIO18:** L298N IN1. **GPIO21:** L298N IN2.
- **5V pin:** from V5. **GND:** to GND. **3V3 pin:** output to V3V3.

Pins to avoid: GPIO0 (boot), GPIO3 (strapping), GPIO19 and GPIO20 (USB), GPIO26 to GPIO32 (internal flash), GPIO45 and GPIO46 (strapping).

Analog rule: keep all analog readings on GPIO1 to GPIO10 (ADC1). The other analog bank stops working while Wi-Fi is on.

Optional battery sensing (not drawn in the diagrams): R12 (100k) from VBAT to node VB_SENSE, R13 (22k) from VB_SENSE to GND, C8 (100 nF) across R13, VB_SENSE to U6.GPIO3. Scale factor 5.55 (12.6 V reads 2.27 V). GPIO3 is a strapping pin; it is fine as a high-impedance input after boot. If you see boot problems, move this to GPIO9 and drop the I2C sensor.

8. Calibration and scaling

- **ADC setup:** set 11 dB attenuation and read with analogReadMilliVolts(). Average 64 samples per channel.
- **Panel voltage:** V = Vadc x 11 (divider 10k / 110k). A 21.6 V panel reads 1.96 V.
- **Panel current:** the ACS712-30A outputs 2.5 V plus 0.066 V per amp. The divider (10k / 14.7k) multiplies by 0.680. So OUT = Vadc / 0.680, then I = (OUT - Vzero) / 0.066.
- **Vzero:** measure at zero current (at night, or with K2 on the open leg) and store it. It is about half the 5 V rail.
- **Reference cell:** I_ref = Vadc / 10 ohm. It is proportional to sunlight.
- **Calibrate against the multimeter:** record real versus ADC values at two points and store scale factors. Expect roughly 5 to 10 percent error at 1 A.
- **Power:** P = V x I. Soiling index SI = (I_A / I_ref) divided by the same ratio right after the last clean.
- **DS18B20:** 12-bit resolution, 750 ms conversion. A reading of -127 or +85 C means a sensor fault. **DHT22:** read no faster than every 2 s.

9. Firmware framework

Modules

- **hal:** pin definitions and output setup (set relay pins HIGH before enabling them, to avoid a click at boot).
- **sensors:** oversampled ADC, DS18B20, DHT22, debounced end stops.
- **calibration:** scale factors and Vzero in non-volatile storage.
- **control:** the state machine below.
- **drivers:** pump relay, mode relay, motor direction and PWM.
- **safety:** interlocks and watchdog.
- **comms:** Wi-Fi, NTP time, dashboard upload, command handling.
- **logger:** timestamped records; a ring buffer of the last 500 entries in flash if Wi-Fi is down.

State machine

- **BOOT:** self-test (sensors sane, pack voltage at least 9.5 V if sensed), then HOMING.
- **HOMING:** drive up at 60 percent PWM until U13 is active. Time-out 30 s leads to FAULT_HOME.
- **IDLE:** read every 5 s; evaluate triggers.
- **COOLING:** entered when Panel A is above 45 C for 3 samples, no rain, level OK, lockout expired. K1 on. Exit below 38 C or after 60 s, or on rain or low level. Then lockout 120 s.
- **CLEANING:** entered when SI stays below 0.90 for 10 minutes with irradiance above 300 W/m2, no rain, level OK; or on schedule (for example 07:00 and 17:00); or on a manual command.
  - PRE_SPRAY: K1 on for 2 s.
  - WIPE_DOWN: motor forward until U14 is active (time-out 25 s).
  - WIPE_UP: motor reverse until U13 is active (time-out 25 s).
  - PARK: motor off, K1 off, record the new baseline.
- **MEASURE_MODE:** every 30 minutes K2 switches Panel A to RL1 for 60 s so A and B run identical loads. Log those rows with a mode flag.
- **FAULT:** all outputs off, fault code logged. Rain and level faults clear on their own; jam, sensor and over-temperature faults need a manual reset.

Interlocks: the pump never turns on unless the level sensor is OK. The motor never drives when both end stops are active. ENA goes to 0 on any fault. The watchdog resets the ESP32 if the loop stalls.

Log record: timestamp, state, VA, IA, VB, IB, power A and B, temperature A, B and tank, ambient temperature and humidity, VREF, SI, pump on-time, motor on-time, mode flag, fault code.

10. Harness and connectors

- **J1:** XT60 battery. **TB1:** 2-way screw terminal, Panel A in. **TB2:** Panel B in. **TB3:** pump. **TB4:** motor. **TB5:** 2-pin JST for the reference cell.
- **H1:** 3-pin JST-XH for the DS18B20 bus. **H2:** DHT22. **H3:** rain. **H4:** level. **H5:** top stop. **H6:** bottom stop. **H7:** 4-pin I2C.
- **Wire colours:** red VBAT, black GND, orange V5, purple V3V3, yellow analog signals, green digital signals, blue control lines to the relay and motor driver.
- **Label** both ends of every wire with its net name.
- **Routing:** run power and motor wires away from analog sense wires. Use cable glands where wires enter the electronics box and add strain relief.

11. Mechanical and electrical interfaces

- **Electronics box:** mount above the splash zone, with cable glands facing down. Everything in section 5 and 6 sits inside, except the panels, pump, tank, sensors and motor.
- **DS18B20 on the panel back:** thermal paste plus aluminium foil tape, cable secured to the frame.
- **End stops:** IR sensors in the rod clamps, facing the carriage block end face about 10 to 30 mm away. Adjust the module's potentiometer until it triggers reliably.
- **Motor and idler wiring:** the motor cable runs along the frame, never across the belt path.
- **Rain sensor:** a pole beside Panel A, tilted like the panel.
- **Level sensor:** on the tank wall or lid; keep the module itself dry.

12. Bring-up and test procedure

Work in this order. Stop and fix any step that does not match.

1. **No battery, no panel:** with the multimeter, check resistance between VBAT and GND is above 1 k ohm (no short), and between V5 and GND is above 1 k ohm.

2. **Pack:** measure each cell (3.0 to 4.2 V, within 0.05 V of each other). Pack total 9.0 to 12.6 V.
3. **Buck only:** connect the pack to U3 only and set the output to 5.00 V plus or minus 0.05 V.
4. **ESP32:** connect V5 to U6.5V. The board powers up; confirm the 3V3 pin reads 3.3 V.
5. **Sensors:** add the DS18B20, DHT22, rain, level and reference cell. Check each reading in the serial monitor.
6. **Panel voltage:** connect Panel A with a multimeter in parallel. The ADC value times 11 should match the multimeter within 2 percent.
7. **Panel current:** with the panel connected through U4 and a load, compare with the multimeter in series. Store the zero offset.
8. **Relays:** with the pump disconnected, toggle K1 and K2 and listen for clicks. Then connect the pump to a bucket.
9. **Motor:** with the belt off, check forward and reverse, then test the end stops by hand.
10. **Full dry run:** assemble the carriage, run one cleaning cycle without water, then with water.
11. **Charger:** connect U1 with the pack and watch the charge current; the pack must not exceed 12.6 V.
12. **Weatherproof** the box last.

13. Failure modes and protections

- **Pump runs dry:** level interlock and the F4 fuse.
- **Carriage jams:** motion time-out, F5 fuse, and FAULT_JAM.
- **Sensor failure (-127 or +85 C, no DHT reply):** fault logged, automatic cooling disabled.
- **Relay clicks at boot:** set pins HIGH before enabling, or drive the relay open-drain.
- **Wi-Fi loss:** local rules keep running; logs buffer in flash.
- **Brown-out when the pump starts:** C3 on V5, the watchdog, and the BMS low-voltage cut.
- **Reverse polarity or back-feed at night:** D1 protects the charging path. Fit a reverse-polarity diode or keyed connectors on the battery lead.
- **Short circuit:** F1 to F5 per branch and the BMS.
- **Overcharge:** charger set to the Li-ion 3S profile plus the BMS.

14. Corrections to earlier values in the chat

- **Dummy loads** are 15 ohm 25 W (not 10 W). At the panel's maximum power point it dissipates close to 20 W.
- **Blocking diode** is SR560 or 1N5822 (not 1N5819, which is rated only 1 A).
- **Panel fuses** F1 and F2 are 3 A (not 5 A).
- **VBAT** feeds loads directly from the BMS output through S1 and F3; the charger's LOAD terminals are unused.
- **K2** now uses the spare relay for measure mode.
- **Added:** fuses F4 and F5, and capacitor C3.

15. Verify against your actual modules

- **L298N:** jumper names and the ENA pin position.
- **Relay module:** whether it is active-low and whether its input works at 3.3 V.
- **ACS712 module:** pin order is usually VCC, OUT, GND; check the silkscreen.
- **LM2596:** the trimmer turns many times; set the output with the multimeter, never by feel.
- **Charge controller and BMS:** terminal labels and the Li-ion mode.
- **DS18B20 wire colours** vary by seller.
- **Optical level sensor:** supply voltage and output level.
- **IR end-stop modules:** whether the output is high or low when triggered.
- **ESP32-S2 board:** use the GPIO numbers on its silkscreen, and do not connect USB and the 5V pin together unless your board has back-feed protection.
- Prices and part availability have not been checked.